

Pyro Sensor

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Problem Definition

Modern smoke detectors do not directly notify emergency responders. The fire department also has limited data about building residents. Our Pyro Sensor enhances safety by immediately alerting users and emergency responders via a mobile app. It also provides critical information, such as the number of residents and pets.

Methodology

The Pyro Sensor comprises three key subsystems:

Hardware

- Sensors: UV, IR, photoelectric, gas; Alarm buzzer.
- Powered with hardwired 120V AC from building or 9V DC from battery.

MCU Firmware

- ESP32-S3 microcontroller: I2C and ADC communication with sensors.
- Sounds buzzer and alerts app once algorithm confirms fire.

Android Application

- Users create accounts, set up workspaces, and add sensors displayed in real time.
- Sends alert notifications to app and central dispatcher.
- Generates unique access codes for firefighter access to sensor and user/building info.



Figure 1. Pyro Sensor node

Engineering Analysis

The Pyro Sensor, either on wall or battery power, uses a multi-sensor algorithm for fire detection. On startup, Bluetooth Low Energy connects the device to Google Firebase, enabling two-way communication with the app.

- A. The Pyro Sensor runs on a 120V AC outlet and seamlessly switches to the 9V backup battery during outages without interrupting operation. Figure 2 compares its power consumption to traditional fire alarms.

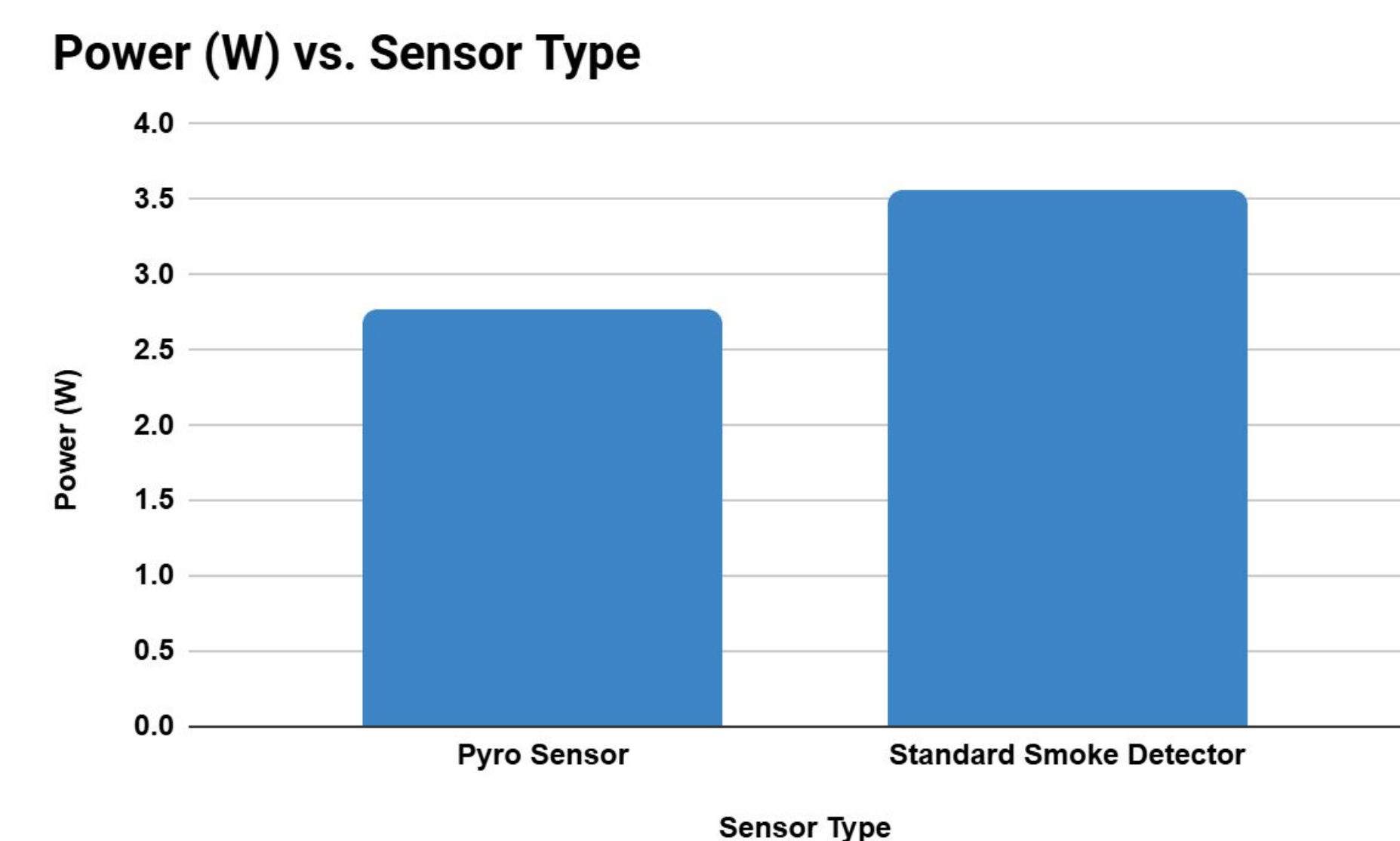


Figure 2. Peak Power Consumption Comparison

- B. The fire detection algorithm (Figure 3) detects flaming and smoldering fires while avoiding false alarms, such as steam.

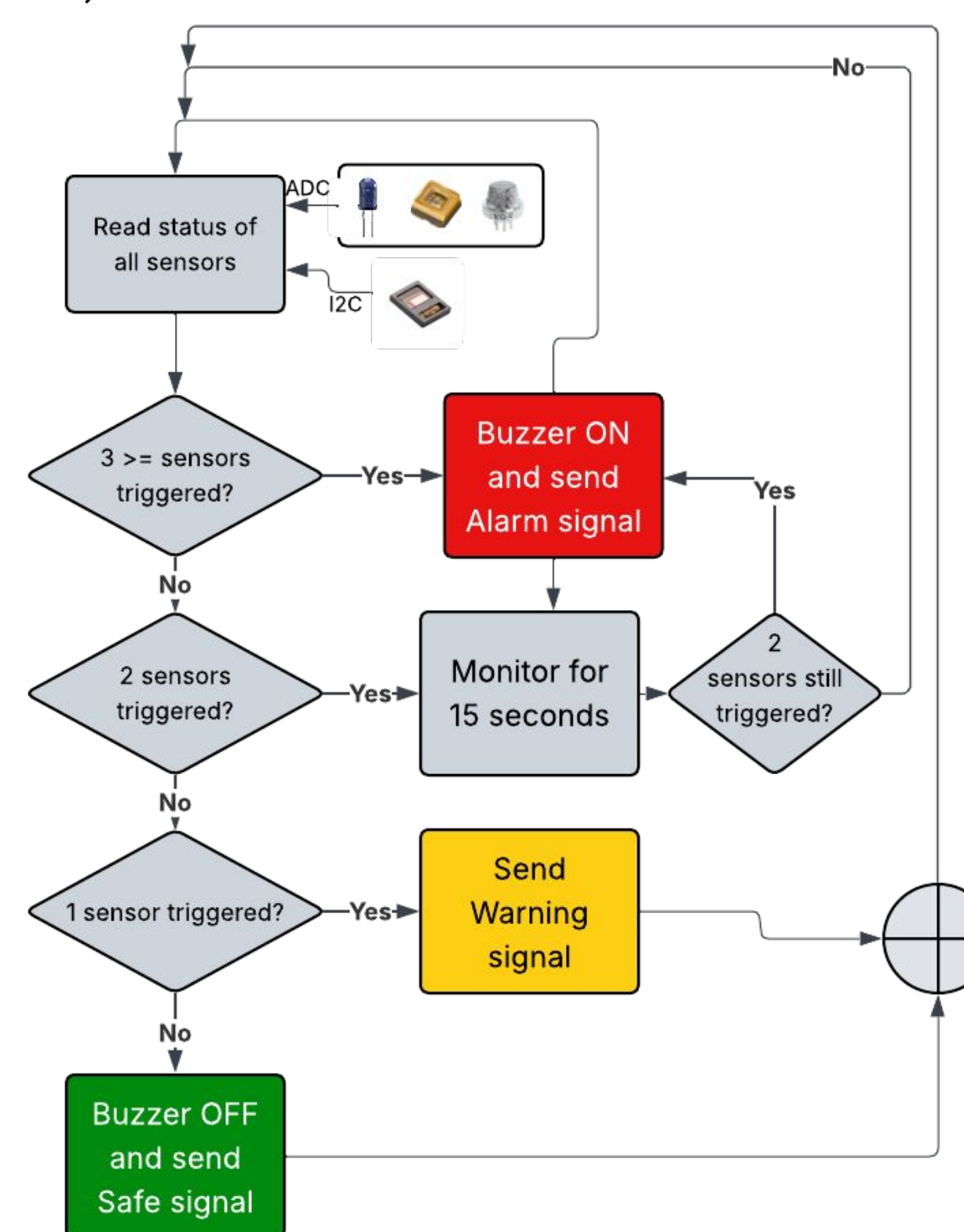


Figure 3. Fire Detection Logic

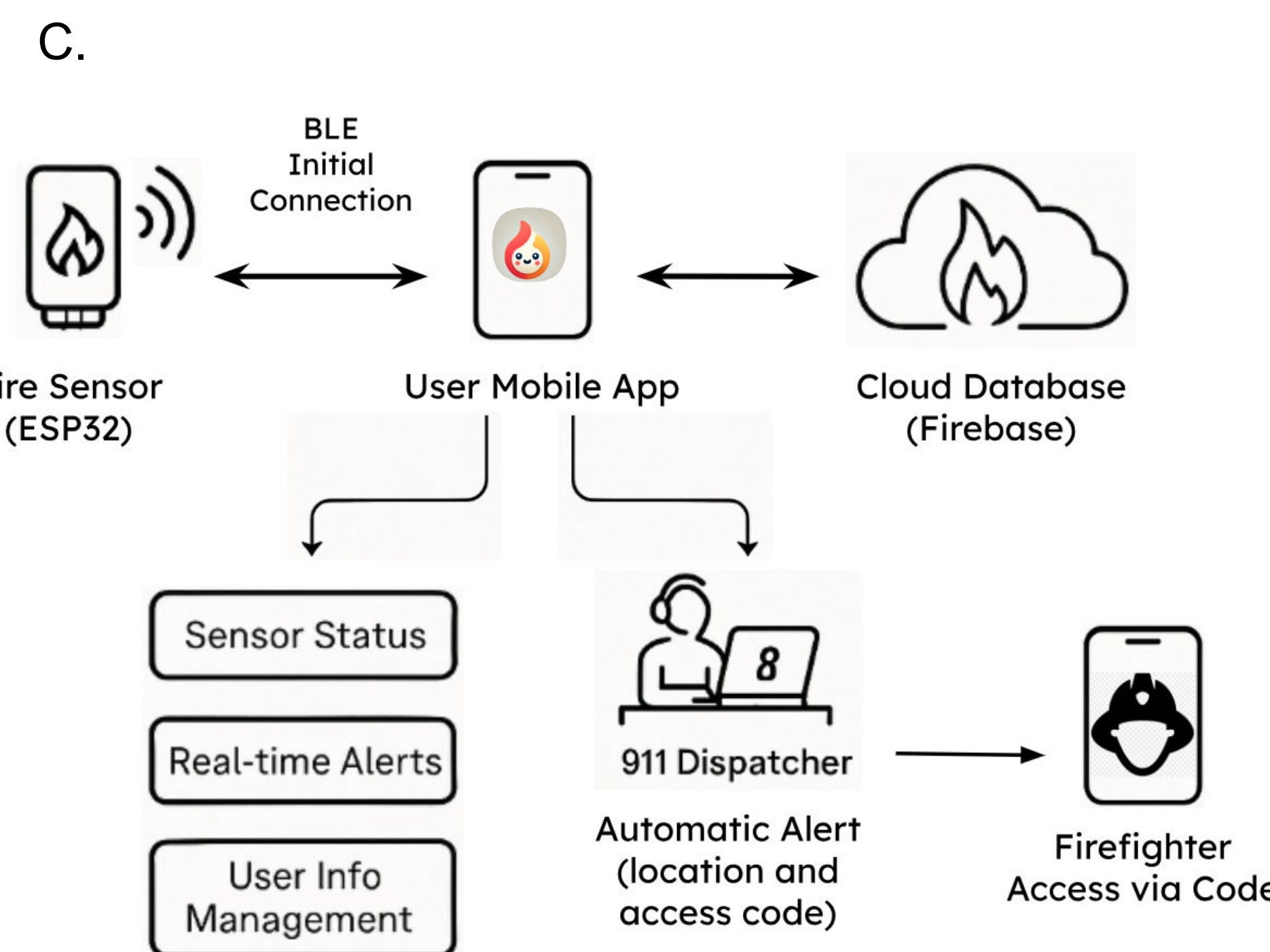


Figure 4. Mobile Application System Flow

Outcomes

The device sends real-time data to Firebase with 0.6s latency, which the app fetches and displays with a user-friendly GUI on the home screen.

Additionally, the user receives an audible notification in case of a fire event.

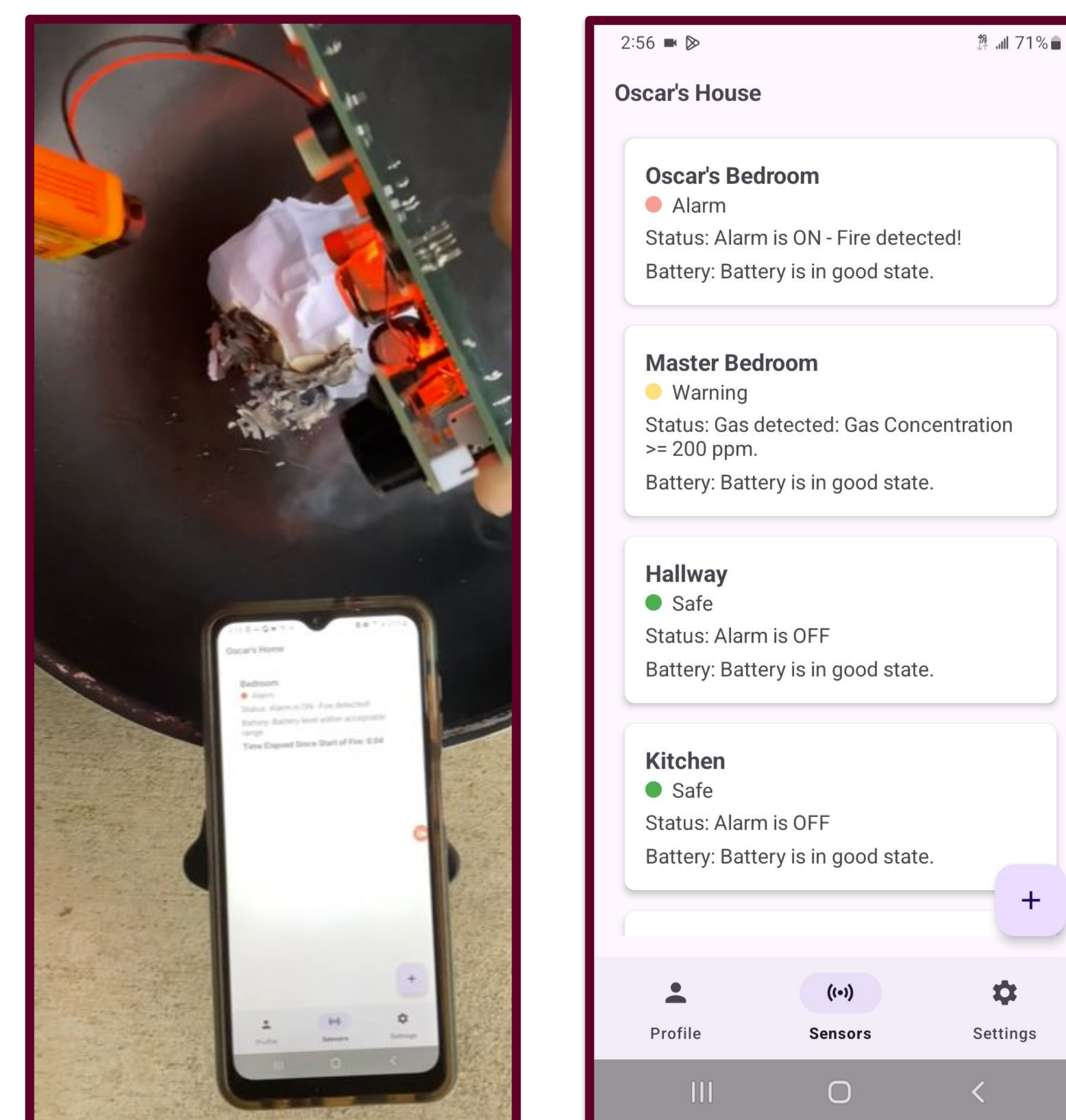


Figure 5 & 6. Demo Run & App Sensor Display

Impact

- By enabling early detection and faster response, the system can help minimize property damage and protect lives during fire emergencies. These preventative measures may also lead to reduced insurance premiums for property owners.
- The system provides real-time alerts and room-specific fire or smoke data directly to the user's phone. This helps users quickly identify the exact location of the incident and act before it spreads.
- Firefighters can use a unique access code to instantly retrieve important details about the building, expected occupants, and fire location. This enables faster and more informed emergency response, potentially saving lives.
- Pyro Sensor is ideal for hotels, apartments, and other multi-room properties that want to enhance safety without overhauling their existing infrastructure. The wireless sensor setup makes installation simple and minimally invasive.

References

- National Fire Protection Association, NFPA 72: National Fire Alarm and Signaling Code, 2019. <https://edufire.ir/storage/Library/elam/NFPA%2072-2019.pdf>
- Detector Electronics Corporation, SmokeWatch™ U5015 Explosion Proof Smoke Detector Specifications, Rev. 2.2, 2020. <https://www.det-tronics.com/content/documents/U5015-Specifications-90-1217-2.2.pdf>
- Espressif Systems, ESP-IDF Programming Guide for ESP32-S3, v5.4.1. <https://docs.espressif.com/projects/esp-idf/en/v5.4.1/esp32s3/index.html>
- Analog Devices, ADPD188BI: Integrated Optical Module for Smoke Detection, Data Sheet. <https://www.analog.com/media/en/technical-documentation/data-sheets/adpd188bi.pdf>
- Firebase Documentation. <https://firebase.google.com/docs>
- Twilio Docs – Programmable Messaging API. <https://www.twilio.com/docs>
- Android Developers. Jetpack and AndroidX Libraries. <https://developer.android.com/jetpack>
- Android Developers. Build UI with XML. <https://developer.android.com/guide/topics/ui/declaring-layout>

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